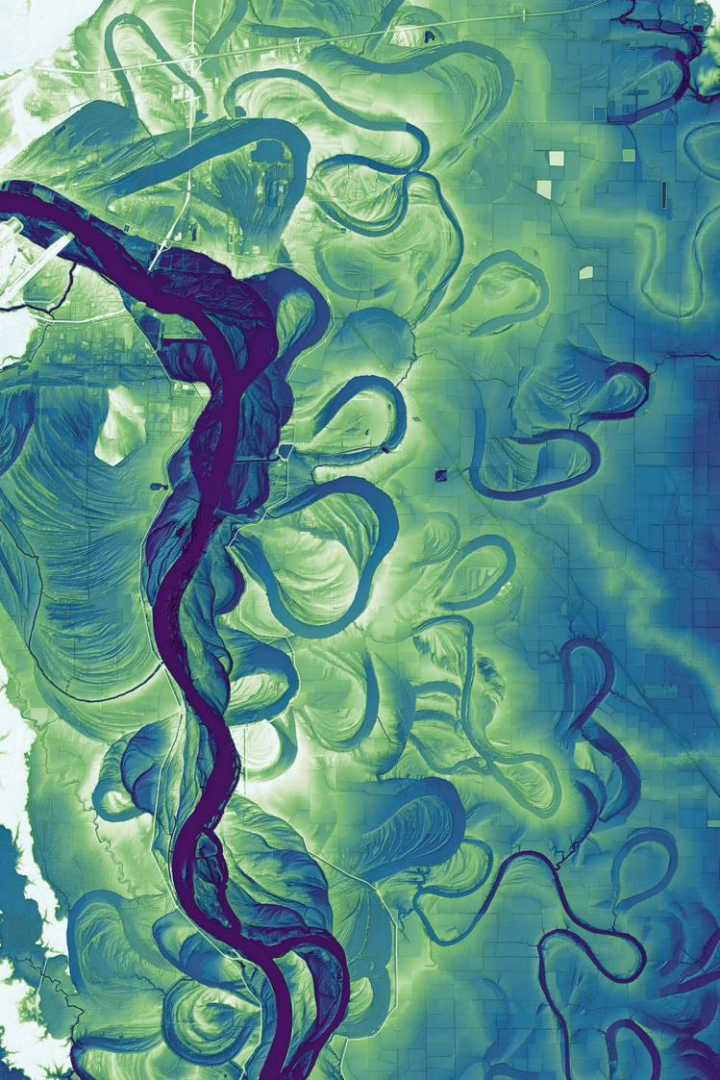




Übung II - Geologische Methoden: Vertiefung (BSc-GV05)

Introduction to GIS and Digital Cartography

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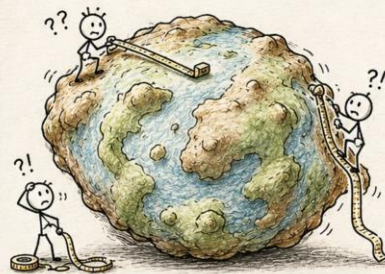
Lesson 2: Coordinates and Projections

Translating the 3D globe into a 2D surface

What is a Datum?

- The physical Earth is bumpy and irregular, making it impossible to use for precise coordinate math.
- A **Datum** solves this by using a smooth mathematical model: an **oblate spheroid** (flattened at the poles, bulging at the equator) to approximate the Earth's shape.
- This smooth surface provides a standardized starting point for assigning horizontal coordinates (Latitude and Longitude).
- Two key types: Global datums (e.g., **WGS84**, which finds the best fit for the whole planet) and Local datums (optimized to fit specific regions perfectly).

1. The Irregular Reality: The Geoid



The Earth's actual shape is extremely irregular and complex.

Impossible to use simple 2D mapping math here!

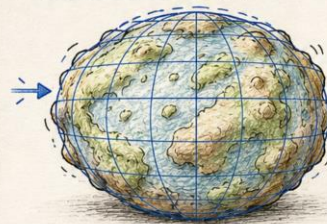
2. The Mathematical Model: Spheroid



A perfectly predictable, smooth mathematical shape.

Easy to calculate simple horizontal coordinates (Latitude & Longitude) on this smooth shape!

3. A Global Datum (Tying the Math to Reality)



WGS84 (A Global Datum) precisely aligns the smooth spheroid reference to best fit the entire lumpy Geoid.

Allows us to use predictable spheroid math to assign accurate global coordinates everywhere!



In short: We use a smooth math model (spheroid) + a global datum (WGS84) so we can speak the same coordinate language anywhere on Earth!



GCS: Measuring a 3D World

A GCS maps the spherical Earth using angular units (degrees).

Latitude

Measures angular distance north or south of the Equator (0° to $\pm 90^\circ$).

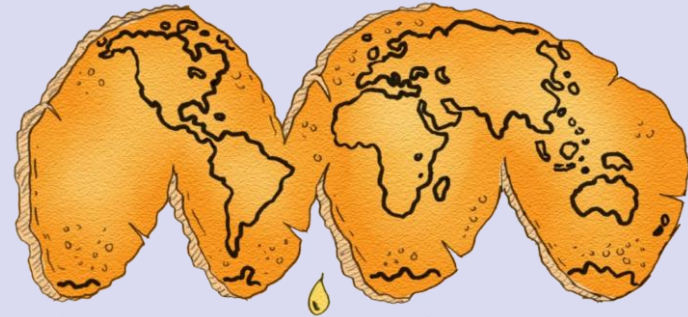
Longitude

Measures angular distance east or west of the Prime Meridian (0° to $\pm 180^\circ$).

- ☐ Coordinates can be expressed in Degrees-Minutes-Seconds (DMS) or Decimal Degrees (DD) — e.g. $51^\circ 30' 26''\text{N} = 51.5072^\circ\text{N}$.

How do you flatten a sphere?

- A spherical surface cannot be mapped onto a flat plane without distortion.
- Flattening requires tearing, stretching, or compressing the surface.
- This is why any 2D map of a globe must sacrifice perfect shape, area, or distance.



Choose what you want to preserve

Every cartographer must choose to preserve at least one of following four key geometric properties.

Shape

Ensure local angles and features look exactly as they do in the real world by distorting area. e.g., Mercator

Area

Ensure all landmasses are sized proportionately to one another by stretching or compressing the shapes

Distance

A map can preserve accurate distance from one specific point (such as the North Pole) to all other points on the map, but this causes distortions in shape and area the further you move from the center

Direction

A map can preserve true direction from a center point to all other points, but this causes distortions in shape and area the further you move from the center

PCS: Translating 3D to 2D

- A PCS translates the 3D angular measurements of a GCS onto a flat 2D grid.
- Uses linear units (metres or feet) rather than angular degrees.
- Enables localized spatial calculations: area, distance, and direction can be measured accurately within the projection's zone.
- Key components: Origin point, X-axis (Easting), Y-axis (Northing), False Easting/Northing to avoid negative values.

Step 1

GCS (Angular)

Geographic coordinates define locations on a curved surface.

Step 2

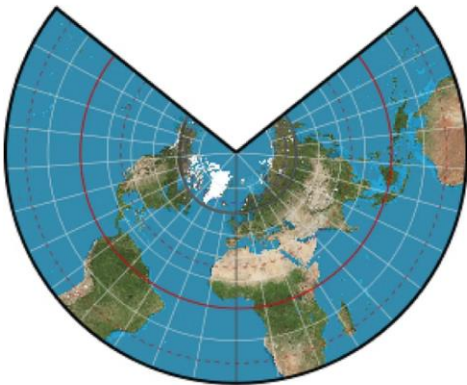
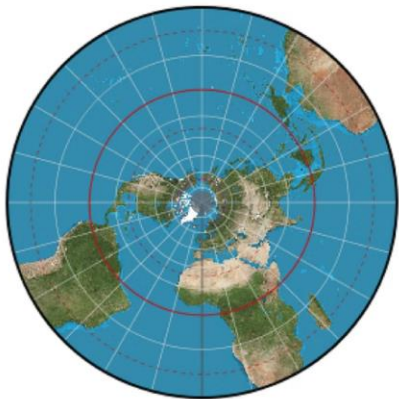
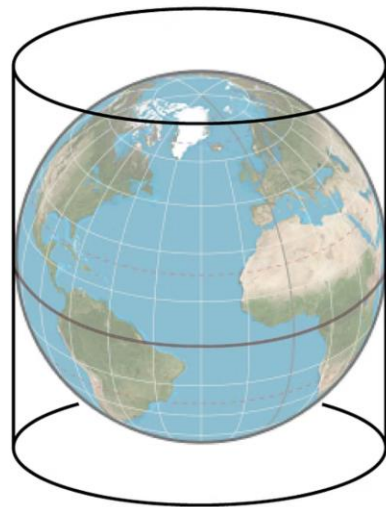
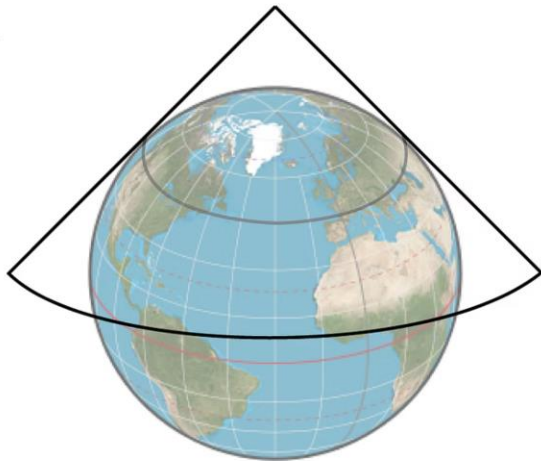
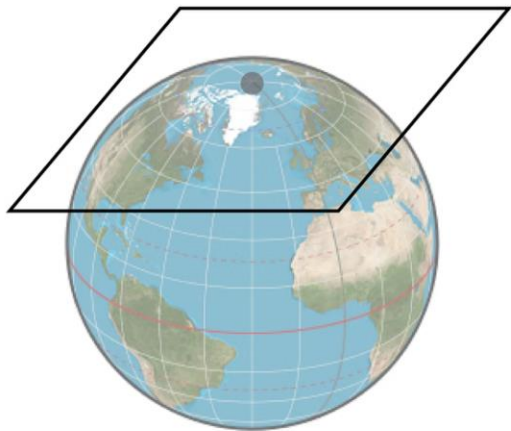
Projection Method

Mathematical formulas transform the spherical surface to a plane.

Step 3

PCS (Linear/Cartesian)

Projected coordinates use X,Y values on a flat grid for precise measurement.



The Projective Surfaces

The three primary developable surfaces used to flatten the Earth in a PCS.



Planar (Azimuthal)

A flat plane is placed tangent to the globe at a single point. Best for polar regions. Example: Stereographic.



Conical

A cone is placed over the globe; the surface is unrolled. Best for mid-latitude regions. Example: Albers Equal-Area.



Cylindrical

A cylinder is wrapped around the globe; the surface is unrolled into a flat plane. Best for equatorial regions. Example: Mercator.

Standardising Coordinate Systems

The EPSG (European Petroleum Survey Group) Registry is a standardised database of coordinate reference systems used globally in GIS software.

An EPSG code is a unique numeric identifier — a "passport" — that encodes the full specification of a coordinate system: datum + projection method + units. It ensures that GIS software can unambiguously interpret the spatial geometry of any dataset.

4326

WGS84 Geographic

(Used by Google Earth)

3857

Web Mercator

(Used by Google Maps)

25832

UTM Zone 32N

(Used in Germany)



Always check the EPSG code of your dataset before performing any spatial analysis.